

TECHNOVA CUSTOMER SUPPORT CHATBOT

Production-Ready Chatbot Evaluation Report

1. Executive Summary

This project implements a stateful TechNova customer-support chatbot using LangGraph, an NVIDIA-hosted LLM, tool calling, conversation memory, and LangSmith observability. The chatbot supports order status, returns, warranty, and shipping questions.

LangSmith was used to trace graph execution, LLM calls, tools, latency, metadata, failures, evaluation experiments, and feedback. The optimized configuration reduced average latency from 9.34 seconds to 5.33 seconds and P95 latency from 42.47 seconds to 18.48 seconds. Both tested models achieved 100% correctness on the 10-example LangSmith evaluation dataset. The 120B model was fastest in the tested benchmark, while the 30B model provides a lower-cost alternative under the assignment's hypothetical pricing assumptions.

2. System Architecture

Flow: User → LangGraph Agent → NVIDIA-hosted LLM → get_order_status / get_policy → Tool Result → Final Response → User.

Conversation state uses MessagesState and MemorySaver, with thread_id preserving context across turns. A custom clean_user_input helper is also traced.

3. Latency and Model Comparison

Metric	Baseline	Optimized	Improvement
Average	9.34 s	5.33 s	42.9%
P50	4.70 s	2.65 s	43.6%
P95	42.47 s	18.48 s	56.5%
Maximum	55.91 s	28.27 s	49.4%

Model	Average	P50	P95	Max	Quality
30B	6.71 s	3.20 s	16.94 s	18.53 s	5/5
120B	1.30 s	1.14 s	2.29 s	2.82 s	5/5

The 120B model recorded substantially lower latency in the tested comparison while matching the 30B model's 100% LangSmith correctness result. Quality is represented as 5/5 based on the observed 10/10 correctness evaluation. Representative streaming measurement: 30B TTFT 0.554 s; 120B TTFT 0.356 s.

4. Token Usage and Cost Analysis

In a 10-turn full-history conversation, input tokens totaled 8,178, output tokens 1,722, and total tokens 9,900. Recent-history optimization reduced input tokens by 77.1% and total tokens by 62.6%.

Metric	30B	120B
10-turn cost	\$0.001980	\$0.008910
20-question cost	\$0.004103	\$0.018238
10K/day	\$19.80/day	\$89.10/day
Monthly	\$594	\$2,673

Costs use the assignment's hypothetical rates of approximately \$0.20/M input/output tokens for the smaller model and \$0.90/M for the larger model. These are exercise estimates, not current NVIDIA commercial pricing.

5. Fault Tolerance and Error Analysis

Five scenarios were tested: invalid model with fallback, invalid/missing API key, timeout with retry/backoff, tool failure, and bad input validation. The chatbot returned user-friendly errors instead of exposing internal technical details.

Fault-injection benchmark	Result
Total requests	20
Successful	12
Failed	8
Error rate	40%

Injected types: 2 each: invalid model, API key, timeout, tool failure, bad input. The 40% error rate is a deliberate fault-injection benchmark and is not a normal production error-rate estimate.

6. LangSmith Evaluation

Dataset: technova-customer-support-eval. It contains 10 customer-support questions covering orders, policies, delivery, free-shipping threshold, and an unknown-order case. An LLM-as-judge correctness evaluator was run against both models.

Model	Examples	Correct	Accuracy
30B	10	10	100%
120B	10	10	100%

Five positive user_rating feedback entries were also added to LangSmith runs.

7. Deployment Recommendation

For the tested workload, the 120B model is recommended when response speed and quality are the primary priorities. The 30B model is recommended for cost-sensitive workloads because it also achieved 100% correctness. A fallback model, retry/backoff handling, and conversation-history optimization should be retained in production.

8. Top 3 Production Risks and Alert Thresholds

1. Latency spikes: investigate when P95 exceeds 10 seconds; high-priority alert above 20 seconds.
2. API/model failures: investigate above 5% error rate and escalate above 10% over a rolling production window.
3. Token growth: monitor average input tokens, daily consumption, and projected monthly cost; apply history trimming or summarization when needed.

9. Conclusion

The TechNova chatbot demonstrates a stateful, tool-using and observable customer-support architecture with measurable latency, token, cost, resilience, and evaluation characteristics. The optimized configuration materially improved latency, while both tested models achieved full correctness on the evaluation dataset. The main production priorities are latency monitoring, failure handling, and control of token growth.

Implementation note: The working implementation uses ChatOpenAI against NVIDIA's OpenAI-compatible endpoint rather than the ChatNVIDIA class because the selected Nemotron model was not recognized by the installed ChatNVIDIA catalog. The NVIDIA-hosted model and LangSmith observability requirements were retained.

10. Screenshots and Observability Evidence

Figure 1. LangSmith Agent & Tool Tracing. The trace shows the LangGraph execution, NVIDIA-hosted model call, tool invocation for TN1001, and returned order information.

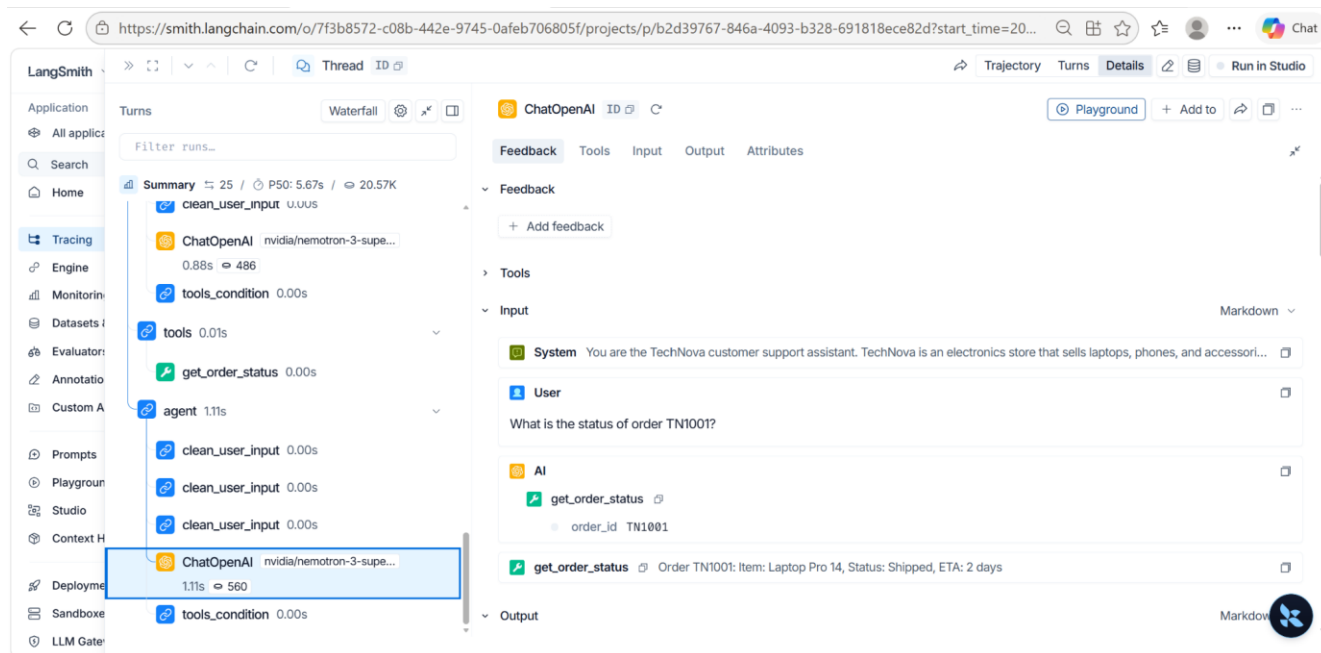


Figure 2. Streamlit Cloud — Chatbot & User Feedback. The deployed TechNova chatbot shows a successful TN1001 order lookup, the model used, and the Helpful/Not helpful feedback mechanism with confirmation.

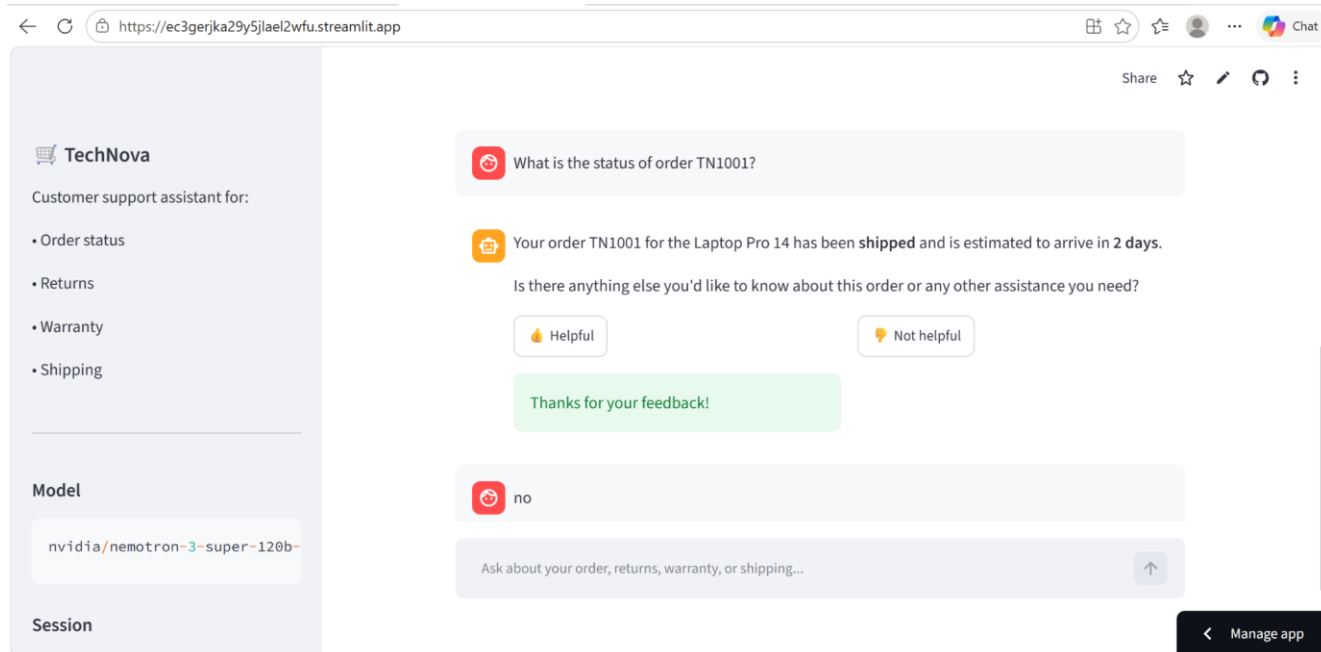
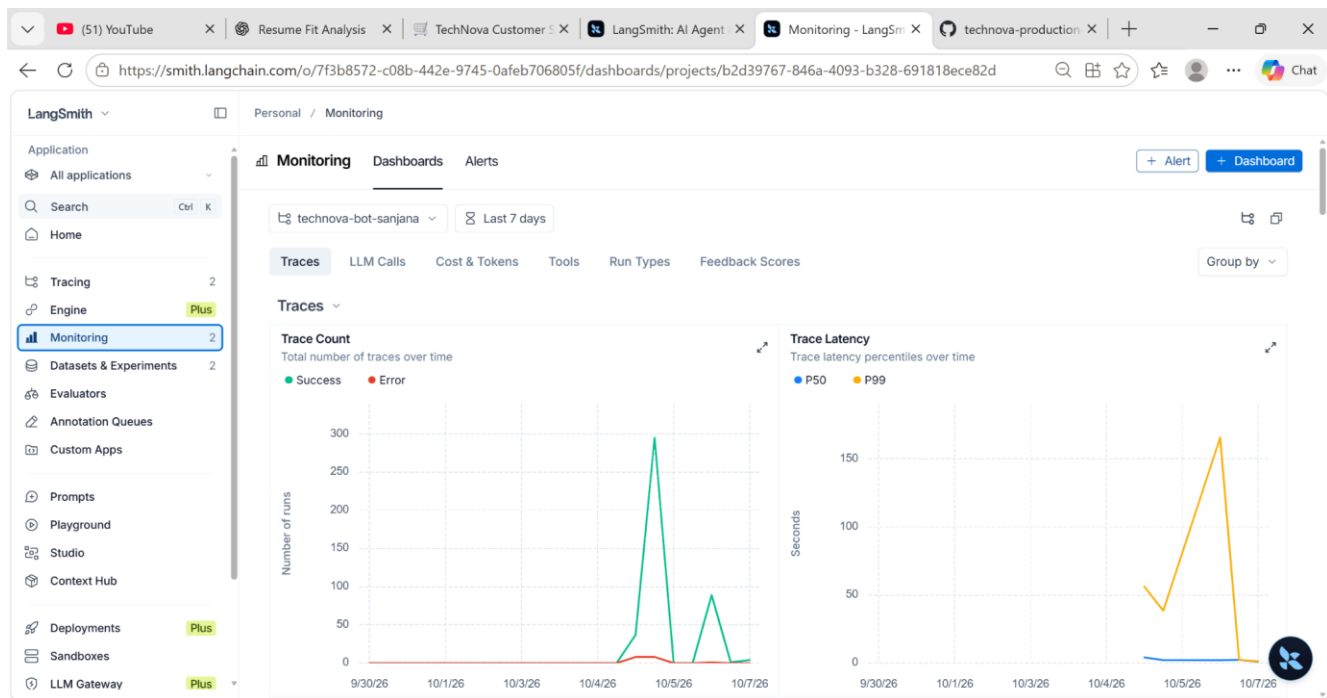


Figure 3. LangSmith Production Monitoring Dashboard

The monitoring view provides production observability through trace count, success/error runs, and latency percentiles (P50/P99), supporting the report's latency and resilience monitoring requirements.



These screenshots provide visual evidence of the implemented tracing, deployed chatbot with user feedback, and production monitoring workflow.